
Linked Open Data Application for Movie Knowledge Graph

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Abstract

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1 Introduction

The rapid growth of digital media has produced a vast amount of structured information about movies, actors, directors, production companies, genres, and ratings. Platforms such as TMDB and IMDb maintain millions of records, and this data is widely used for search, recommendation, and analytics. However, it is often scattered across isolated databases and tabular datasets, each with its own schema, identifiers, and access method. As a result, it is difficult to query across sources, to connect semantically related entities or to ask questions that span several entity types. Knowledge graphs address this problem by representing entities and their relationships in an interconnected graph structure, supporting integration, exploration, and reasoning across heterogeneous information sources [Hogan et al., 2021]. This perspective is consistent with the Linked Data principles of using dereferenceable identifiers and links to related resources [Berners-Lee, 2006].

This work develops a Linked Open Data (LOD) application centered on the movie domain. It covers movies and related entities, including people, organizations, genres, countries, languages, and ratings, together with relationships such as cast membership, directorship, production affiliation, and publication date. The resulting knowledge graph is designed as a reusable and interoperable semantic resource, while remaining focused on a representative subset of the domain that can be modeled consistently using RDF, OWL, and Schema.org vocabulary [Schema.org, 2025, W3C RDF Working Group, 2014]. Reusing Schema.org terms rather than inventing new ones lowers the barrier for other applications to interpret the data.

We demonstrate a complete pipeline for building a movie knowledge graph in accordance with LOD principles. As illustrated in Figure 1, the proposed workflow consists of six stages: (1) defining an ontology for the movie domain; (2) collecting relevant movie data from TMDB; (3) transforming the collected CSV data into RDF in Turtle notation using stable URIs; (4) building an OWL ontology that defines the domain classes, properties, and restrictions; (5) establishing links to DBpedia and Wikidata to improve interoperability and achieve a higher level of Linked Open Data integration; and (6) providing a SPARQL interface through Apache Jena Fuseki for querying the resulting graph [W3C SPARQL Working Group, 2013]. As an optional stage, we provide a static web application, hosted on GitHub Pages in the style of DBpedia, to support browsing and exploration of the generated data.

The remainder of this report is organized as follows. Section 2 details our method, defines the ontology for the movie domain, and specifies the evaluation protocol. Section 3 presents the data collection and transformation. Section 4 introduces the ontology construction and entity linking.

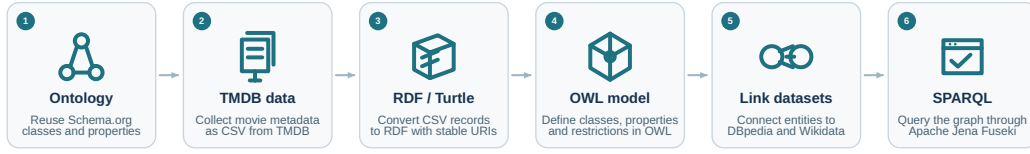


Figure 1: Overview of our method.

Section 5 describes the evaluation. Section 6 discusses limitations and potential future work. Finally, section 7 concludes.

2 Methodology

Figure 1 summarizes the proposed pipeline for constructing and exposing a movie knowledge graph. Movie data are collected from TMDB and stored in CSV files. These tabular records are transformed into Turtle RDF using stable resource URIs. The OWL ontology then adds class, property, and restriction semantics and supports OWL-RL inference. Movie entities are linked to DBpedia and Wikidata using identifiers, titles, release years, and related matching criteria. Apache Jena Fuseki provides a browser-accessible SPARQL endpoint for querying the graph. The optional static web application publishes generated resource pages and ontology views for human browsing.

Here, we describe the first step of the workflow (i.e., defining the ontology for the movie domain) and specify the evaluation protocol, which assesses the resulting knowledge graph by executing representative SPARQL queries. The remaining stages are described in their corresponding sections.

Movie ontology We construct the movie ontology by reusing Schema.org classes and properties to represent movies and their related entities. Table 1 lists all Schema.org terms used in the ontology [Schema.org, 2025]. The classes model movies, people, roles, descriptive entities, organizations, and ratings, while the properties describe movie metadata and relationships, including titles, identifiers, release dates, duration, genres, languages, countries, production companies, ratings, directors, actors, and acting roles. This vocabulary provides a shared semantic model while allowing the data to be serialized as RDF and constrained or extended with OWL.

Evaluation protocol The evaluation protocol queries the graph using SPARQL. The query set covers basic retrieval, such as listing movies, people, and directors, as well as multi-condition questions involving ratings, genres, release years, and cast members. Additional queries test the value of OWL-RL reasoning by asking for inferred relationships, such as a person’s directed movies or a production company’s country. Finally, federated queries retrieve information about a movie from DBpedia or Wikidata through its established links. Queries are executed against the local graph, with reasoning enabled when required, and against the corresponding remote endpoint for external-source questions.

3 Data collection and transformation

4 Ontology constructing and Entity linking

5 Evaluation

6 Discussion

Limitations

Future work

Table 1: Schema.org classes and properties reused in the movie ontology.

Category	Schema.org terms and use
Classes	<code>Movie</code> (movie); <code>CreativeWork</code> (movie superclass); <code>Person</code> (actor, director, or crew member); <code>Role</code> (reified acting role); <code>Genre</code> (movie genre); <code>Country</code> (movie or company country); <code>Language</code> (movie language); <code>Organization</code> (production company); <code>AggregateRating</code> (movie rating); <code>Rating</code> , <code>Thing</code> , and <code>Intangible</code> (reused superclasses).
Object properties	<code>actor</code> (movie or role to actor); <code>aggregateRating</code> (movie to rating); <code>contributor</code> (movie to non-director crew member); <code>countryOfOrigin</code> (movie to country); <code>director</code> (movie to director); <code>genre</code> (movie to genre); <code>inLanguage</code> (movie to language); <code>location</code> (company to country); <code>productionCompany</code> (movie to company); <code>sameAs</code> (movie to an equivalent external resource).
Datatype properties	<code>abstract</code> (movie synopsis); <code>characterName</code> (character portrayed in a role); <code>datePublished</code> (release date); <code>duration</code> (runtime); <code>identifier</code> (TMDb or IMDb identifier); <code>name</code> (entity name); <code>position</code> (cast order); <code>ratingCount</code> (number of votes); <code>ratingValue</code> (rating value); <code>roleName</code> (crew job title).

7 Conclusion

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